

Continuous-Parameter Edgeworth Theory, Large Deviations, and Quadratic q -Gevrey Regularity at the Fabius–Rvachev Frontier

A repository-audited research report with proofs, conjectures, and reproducible numerics

OpenAI GPT-5.6 Pro

30 August 2026

Abstract

The Fabius function, Rvachev’s compactly supported function up, and their geometric q -deformations sit at an unusually dense intersection of probability, infinite sinc products, automatic sequences, q -series, exact dyadic arithmetic, inverse-function asymptotics, and ultradifferentiable analysis. This report audits the current LaTeX corpus under `Analysis/FabiusFunction/docs` in the PROVEIT repository and develops results selected specifically to avoid duplicating the canonical exposition and its frontier drafts.

The first result closes an explicitly recorded gap in the repository: for the standardized geometric-uniform law with $r = e^{-h} \uparrow 1$, we prove an all-orders Edgeworth expansion in every fixed Schwartz seminorm. Its coefficients are finite Bernoulli–Bell–Hermite polynomials generated by exact standardized cumulants. We derive the first three corrections, uniform CDF expansions, and an all-orders central Cornish–Fisher theory for the inverse distribution. The second group of results identifies two new probability scales. We prove Gaussian moderate deviations and a compact-scale large-deviation principle with speed h^{-1} . The limiting cumulant generating function is

$$\Lambda(s) = \int_0^1 \frac{\log(\sinh(\sqrt{6} su)/(\sqrt{6} su))}{u} du.$$

Its Legendre transform admits an exact saddle parametrization. At the limiting support edge the saddle inversion is governed by W_{-1} , and the rate contains the renormalized constant

$$\gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12},$$

where γ_1 is the first Stieltjes constant. This furnishes a direct bridge between the Gaussian $q \uparrow 1$ regime and the Lambert– W endpoint machinery emphasized elsewhere in the corpus.

Finally, using the repository’s exact Thue–Morse derivative norms, we identify the sharp global Denjoy–Carleman class of up and of the separated-base atomic functions h_a , $a \geq 2$. These functions are not Gevrey of any finite order; their optimal derivative scale is quadratic-exponential, $M_n \asymp a^{n^2/2}$, and the corresponding class is nonquasianalytic. Reproducible Python computations verify the predicted successive error orders and the Lambert– W edge law. The final sections formulate a matched central/moderate/large/endpoint asymptotic program, local symbolic Carleman spectra, and several arithmetic and transseries conjectures.

Novelty convention. “New” in this report means absent from the inspected PROVEIT snapshot and from the source papers used as its baseline. No claim of global literature priority is made without a separate exhaustive bibliographic search. All theorem statements below are proved in the report; conjectures and heuristic extensions are labeled explicitly.

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1 Executive summary and principal contributions

1.1 The geometric-uniform family

Let $(U_n)_{n \geq 0}$ be independent $\text{Unif}[0, 1]$ random variables and let

$$X_r = (1 - r) \sum_{n=0}^{\infty} r^n U_n, \quad 0 < r < 1. \quad (1.1)$$

The weights sum to one, so $X_r \in [0, 1]$. At $r = 1/2$, its distribution function is the Fabius function F , and its density is the affine rescaling [9, 1, 14]

$$F'(x) = 2 \operatorname{up}(2x - 1), \quad 0 < x < 1. \quad (1.2)$$

Thus (1.1) is simultaneously a probability deformation of F , a q -analog with $q = r$, and an infinite-convolution deformation of Rvachev's up.

Put

$$r = e^{-h}, \quad \sigma_r^2 = \frac{1 - r}{12(1 + r)}, \quad Z_h = \frac{X_r - 1/2}{\sigma_r}. \quad (1.3)$$

The report establishes the following package.

Proved frontier results.

- (I) **All-orders continuous-parameter Edgeworth theorem.** The density of Z_h has an expansion to arbitrary order in h , uniformly with polynomial weights and after any fixed number of derivatives; see [Theorem 5.1](#).
 - (II) **Bernoulli–Bell coefficient calculus.** Every standardized cumulant has an explicit all-orders h -expansion whose coefficients are complete Bell polynomials in Bernoulli numbers; see [Theorem 4.2](#).
 - (III) **Central inverse theory.** The CDF and quantile admit all-orders uniform central expansions. The first two Cornish–Fisher terms are explicit; see [Theorem 6.2](#).
 - (IV) **Moderate deviations.** Every scale $b_h \rightarrow \infty$ with $hb_h^2 \rightarrow 0$ has the Gaussian rate $x^2/2$; see [Theorem 7.1](#).
 - (V) **Compact-scale large deviations.** $Y_h = \sqrt{h} Z_h$ satisfies an LDP with speed h^{-1} , compact limiting support $[-\sqrt{6}, \sqrt{6}]$, and an explicit integral cumulant generator; see [Theorem 7.2](#).
 - (VI) **Lambert– W support-edge law.** The large-deviation rate has an exact saddle parametrization and a W_{-1} edge expansion containing a Stieltjes constant; see [Theorems 7.3 and 8.2](#).
 - (VII) **Sharp quadratic q -Gevrey classification.** The exact Thue–Morse derivative norms place up and h_a , $a \geq 2$, in optimal nonquasianalytic Denjoy–Carleman classes and in no finite Gevrey class; see [Theorems 9.1 and 9.3](#).
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The central Edgeworth theorem resolves a program that the repository itself marks as open: its discrete finite-level analog was completed, while the continuous parameter limit $a \downarrow 1$, equivalently $r \uparrow 1$, remained open. A companion frontier draft records the first fixed-frequency quartic correction and conjectures precisely the weighted, differentiated, all-orders statement proved here [15, 16].

1.2 Why the new scales are natural

The standard CLT scale keeps $Z_h = O(1)$. The standardized support, however, is

$$|Z_h| \leq A_h, \quad A_h = \frac{1}{2\sigma_r} = \sqrt{3 \frac{1+r}{1-r}} = \sqrt{3 \coth(h/2)} \sim \sqrt{\frac{6}{h}}. \quad (1.4)$$

Hence $Y_h = \sqrt{h}Z_h$ has a nondegenerate compact limiting support. Three nested regimes appear:

$$Z_h = O(1) \longrightarrow Z_h = b_h x, \ 1 \ll b_h \ll h^{-1/2} \longrightarrow Y_h = \sqrt{h}Z_h = O(1).$$

They are governed respectively by Edgeworth expansions, Gaussian moderate deviations, and a nonquadratic compact-support rate function. Letting the compact coordinate approach $\sqrt{6}$ then creates the Lambert– W_{-1} phase. This four-layer structure is the conceptual backbone of the report.

2 Repository audit and nonduplication protocol

2.1 Corpus map

The inspected repository snapshot was the public main branch on 30 August 2026. The live documentation tree contains five canonical LaTeX works, a large thematically reorganized draft corpus, imported source papers, and a frozen historical archive. The audit used the following hierarchy.

Corpus stratum	Role in the nonduplication audit
Fabius_Function_and_Rvachev_Up	Primary synthesis: construction, probability, Fourier products, moments, exact dyadic values, q -specialization, inverse transport, Lambert– W endpoint asymptotics, and many auxiliary transforms.
semi-formalized-research-frontiers	Canonical frontier volume: probabilistic engines, repeated integration, exact dyadic web, Thue–Morse prefix sums, endpoint and inverse asymptotics.
Non_Elementarity_of_the_Fabius_Function	Nowhere-local-elementarity and formalization-backed analytic-locus arguments.
fabius_lean_walkthrough	Lean architecture, definitions, theorem dependencies, and proof-engineering constraints.
FabiusFunction_Mathematical_Glossary	Notation and cross-document terminology.
Draft manifest and raw candidate drafts	Fourier decay, q -Pochhammer and q -binomial inverses, Legendre/Jacobi/Padé systems, total positivity, interpolation on geometric combs, atomic-function general bases, Gaussian boundary limits, reciprocal transforms, and many archived research waves.
Imported papers	Arias de Reyna on the compactly supported function and Fabius arithmetic; Allouche–Shallit on Thue–Morse; lacunary-product papers; a Thue–Morse Dirichlet-series paper; and related source material.
Historical archive	Frozen first-introduction versions grouped by subject. Byte-identical copies already represented in the live tree are deliberately pruned by the repository’s own archive policy.

The imported baseline includes the classical compact-support and arithmetic papers, the standard Thue–Morse survey, lacunary-product work, and a recent Thue–Morse Dirichlet-series paper [1, 2, 3, 4, 5, 18].

The archive and draft manifest are important methodologically: they prevent a recent theorem hidden in a prior wave from being misidentified as new. Historical copies were treated as lineage evidence, not as independent mathematical content.

2.2 Results deliberately excluded as already present

The following substantial themes were excluded from the novelty claims because the corpus already develops them:

- construction, uniqueness, refinement equations, support, symmetry, monotonicity, flatness, and nonanalyticity of F and up;
- probability representations by weighted uniform sums, finite spline approximants, weak convergence, and general absolutely summable weights;
- the geometric q -family at fixed $0 < q < 1$, including negative-parameter dualities, support geometry, plateaux, reciprocal bases, and the first Gaussian correction;
- infinite sinc products, log-Gaussian Fourier envelopes, Poisson summation, moments, Legendre expansions, and exact derivative hierarchies;
- exact dyadic rational formulas, 2-adic valuations, q -binomial identities, and geometric interpolation;
- fixed-parameter endpoint asymptotics with periodic fluctuations and Lambert– W inversion;
- discrete finite-level Edgeworth expansions for the spline approximants;
- exact derivative norms and equimeasurability for separated-base atomic functions h_a , $a \geq 2$.

The new work starts where those results stop. In particular, the corpus states that the continuous-parameter Edgeworth program remains open, and another draft conjectures a complete Bernoulli–Bell–Hermite expansion with polynomial weights and differentiated uniformity. Searches of the canonical and frontier sources found no compact-scale large-deviation principle, no Gärtner–Ellis rate integral, no Cornish–Fisher expansion, and no completed Denjoy–Carleman classification. The exact derivative norms used in Section 9 are therefore inputs from the repository; the functional-class consequences are new deductions.

2.3 Audit limitations and reproducibility

The tree includes reorganized drafts and frozen versions whose text is intentionally duplicated. The audit therefore used the canonical documents as the semantic baseline, the manifest and archive inventory to identify equivalence classes, and raw-source inspection whenever a proposed result was close to an existing topic. This is a content audit, not a claim that every historical byte was independently reread after duplication was established.

All computations in Section 10 are reproduced by

numerics/fabius_q_edgeworth.py.

The archive accompanying this report contains the script, generated CSV and LaTeX tables, and vector/PNG figures.

3 Preliminaries: Fabius, Rvachev, and the geometric q -family

3.1 Random-series identities

Write $V_n = U_n - 1/2$, so that $V_n \sim \text{Unif}[-1/2, 1/2]$. Then

$$X_r - \frac{1}{2} = (1-r) \sum_{n \geq 0} r^n V_n. \quad (3.1)$$

The variance follows immediately:

$$\sigma_r^2 = \frac{(1-r)^2}{12} \sum_{n \geq 0} r^{2n} = \frac{1-r}{12(1+r)}. \quad (3.2)$$

The moment generating function of V_n is $\sinh(t/2)/(t/2)$, while its characteristic function is $\text{sinc}_{\text{rad}}(t/2) = \sin(t/2)/(t/2)$. Hence the standardized characteristic function is

$$\Psi_h(t) = \mathbb{E} e^{itZ_h} = \prod_{n=0}^{\infty} \text{sinc}_{\text{rad}}(a_h r^n t), \quad a_h = \sqrt{3(1-r^2)}, \quad r = e^{-h}. \quad (3.3)$$

The normalization is exact because

$$\sum_{n \geq 0} (a_h r^n)^2 = 3. \quad (3.4)$$

3.2 Smoothness and Fourier conventions

We use

$$\widehat{f}_{\text{ang}}(-t) = \int_{\mathbb{R}} e^{itx} f(x) \, dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \widehat{f}_{\text{ang}}(-t) \, dt. \quad (3.5)$$

For every fixed $h > 0$, the product (3.3) decays faster than every negative power, and therefore Z_h has a compactly supported C^∞ density f_h [11, 1]. A quantitative version is proved in [Lemma 5.2](#).

The probabilists' Hermite polynomials are

$$H_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}, \quad \varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}. \quad (3.6)$$

Thus $\varphi^{(n)} = (-1)^n H_n \varphi$, and Fourier inversion in the positive-exponent convention (3.5) gives

$$\frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} (it)^n e^{-t^2/2} \, dt = H_n(x) \varphi(x). \quad (3.7)$$

3.3 Cumulants of a centered uniform variable

With Bernoulli numbers defined by

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} B_n \frac{z^n}{n!}, \quad (3.8)$$

we have

$$\log \frac{\sinh(z/2)}{z/2} = \sum_{m=1}^{\infty} \frac{B_{2m}}{2m} \frac{z^{2m}}{(2m)!}, \quad |z| < 2\pi. \quad (3.9)$$

Accordingly, the $2m$ -th cumulant of V_n is $B_{2m}/(2m)$, and all odd cumulants vanish.

4 Exact cumulants and the Bernoulli–Bell bridge

Proposition 4.1 (Exact standardized cumulants). *For every $m \geq 2$, the standardized cumulant of order $2m$ of Z_h is*

$$\lambda_{2m}(r) = \frac{B_{2m}}{2m} 12^m \frac{(1-r^2)^m}{1-r^{2m}}. \quad (4.1)$$

Equivalently, for $r = e^{-h}$,

$$\lambda_{2m}(h) = \frac{B_{2m}}{2m} 12^m 2^{m-1} \frac{\sinh^m h}{\sinh(mh)}. \quad (4.2)$$

In particular, $\lambda_{2m}(h) = \mathcal{O}(h^{m-1})$ as $h \downarrow 0$.

Proof. Cumulants add under independent summation and scale homogeneously. From (3.1) and (3.9),

$$\kappa_{2m}(X_r - 1/2) = \frac{B_{2m}}{2m} (1-r)^{2m} \sum_{n \geq 0} r^{2mn} = \frac{B_{2m}}{2m} \frac{(1-r)^{2m}}{1-r^{2m}}.$$

Dividing by σ_r^{2m} , with (3.2), proves (4.1). Since

$$1 - e^{-2h} = 2e^{-h} \sinh h, \quad 1 - e^{-2mh} = 2e^{-mh} \sinh(mh),$$

the exponential factors cancel and give (4.2). Finally $\sinh^m h / \sinh(mh) \sim h^{m-1}/m$. $\square$

The hyperbolic representation makes a second layer of structure visible. Set

$$c_k = \frac{2^{2k-1} B_{2k}}{k(2k)!}, \quad d_{m,k} = c_k (m - m^{2k}). \quad (4.3)$$

The classical identity

$$\log \frac{\sinh z}{z} = \sum_{k=1}^{\infty} c_k z^{2k}, \quad |z| < \pi, \quad (4.4)$$

turns every cumulant into an exponential Bell series.

Theorem 4.2 (Bernoulli–Bell expansion of every cumulant). *Let Y_ℓ denote the complete exponential Bell polynomial, characterized by*

$$\exp \left(\sum_{k \geq 1} x_k \frac{z^k}{k!} \right) = \sum_{\ell \geq 0} Y_\ell(x_1, \dots, x_\ell) \frac{z^\ell}{\ell!}. \quad (4.5)$$

For $m \geq 2$, define

$$A_m = \frac{B_{2m}}{2m} 12^m \frac{2^{m-1}}{m}. \quad (4.6)$$

Then, convergently for sufficiently small h ,

$$\lambda_{2m}(h) = A_m h^{m-1} \exp \left(\sum_{k \geq 1} d_{m,k} h^{2k} \right) \quad (4.7)$$

$$= A_m h^{m-1} \sum_{\ell=0}^{\infty} \frac{Y_\ell(1!d_{m,1}, 2!d_{m,2}, \dots, \ell!d_{m,\ell})}{\ell!} h^{2\ell}. \quad (4.8)$$

Thus the complete continuous-parameter cumulant calculus is generated by Bernoulli numbers followed by one Bell-polynomial exponentiation.

Proof. Rewrite (4.2) as

$$\lambda_{2m}(h) = A_m h^{m-1} \left(\frac{\sinh h}{h} \right)^m \frac{mh}{\sinh(mh)}.$$

Taking logarithms of the factor after $A_m h^{m-1}$ and applying (4.4) gives

$$m \log \frac{\sinh h}{h} - \log \frac{\sinh(mh)}{mh} = \sum_{k \geq 1} c_k (m - m^{2k}) h^{2k},$$

which is (4.7). Apply (4.5) with $z = h^2$ and $x_k = k! d_{m,k}$. $\square$

The first cumulants, useful later, are

$$\lambda_4(h) = -\frac{6}{5} \tanh h = -\frac{6}{5} h + \frac{2}{5} h^3 - \frac{4}{25} h^5 + \mathcal{O}(h^7), \quad (4.9)$$

$$\lambda_6(h) = \frac{192}{7} \frac{\sinh^3 h}{\sinh 3h} = \frac{64}{7} h^2 - \frac{64}{7} h^4 + \mathcal{O}(h^6), \quad (4.10)$$

$$\lambda_8(h) = -\frac{3456}{5} \frac{\sinh^4 h}{\sinh 4h} = -\frac{864}{5} h^3 + \frac{1728}{5} h^5 + \mathcal{O}(h^7). \quad (4.11)$$

5 All-orders Edgeworth expansion in Schwartz seminorms

5.1 The exact-cumulant Edgeworth polynomial

For a multi-index $\mathbf{k} = (k_2, \dots, k_{R+1}) \in \mathbb{N}_0^R$, define its Edgeworth weight and Hermite degree by

$$w(\mathbf{k}) = \sum_{m=2}^{R+1} (m-1) k_m, \quad \nu(\mathbf{k}) = \sum_{m=2}^{R+1} m k_m. \quad (5.1)$$

Define

$$\mathcal{E}_{R,h}(x) = \sum_{\substack{\mathbf{k} \in \mathbb{N}_0^R \\ w(\mathbf{k}) \leq R}} H_{2\nu(\mathbf{k})}(x) \prod_{m=2}^{R+1} \frac{1}{k_m!} \left(\frac{\lambda_{2m}(h)}{(2m)!} \right)^{k_m}. \quad (5.2)$$

The zero multi-index contributes 1. The polynomial uses the *exact* cumulants, not their truncated power series. This makes the theorem compact and automatically incorporates many higher-order parameter terms.

For integers $j, K \geq 0$, write

$$\|g\|_{j,K} = \sup_{x \in \mathbb{R}} (1 + |x|)^K \left| g^{(j)}(x) \right|. \quad (5.3)$$

Theorem 5.1 (All-orders continuous-parameter Edgeworth expansion). *Fix integers $R, j, K \geq 0$. There exist $h_0 > 0$ and $C_{R,j,K} < \infty$ such that, for $0 < h \leq h_0$,*

$$\|f_h - \varphi \mathcal{E}_{R,h}\|_{j,K} \leq C_{R,j,K} h^{R+1}. \quad (5.4)$$

Consequently f_h has a complete asymptotic expansion in the Schwartz topology, and every coefficient is an explicit Bernoulli–Bell–Hermite polynomial.

Relative to the classical Edgeworth framework [6, 13], this theorem strengthens the usual local statement in three ways: it is uniform on all of $\mathbb{R}$, survives any fixed number of spatial derivatives, and carries arbitrary fixed polynomial weights.

5.2 Global sinc-product bounds

The proof starts with a uniform frequency envelope.

Lemma 5.2 (Global sinc envelope). *There are universal constants $c, C > 0$ such that*

$$|\operatorname{sinc}_{\text{rad}} u| \leq \exp(-c \min(u^2, 1)), \quad u \in \mathbb{R}. \quad (5.5)$$

Let

$$S_h(t) = \sum_{n \geq 0} \min((a_h r^n t)^2, 1). \quad (5.6)$$

Then

$$|\Psi_h(t)| \leq e^{-c S_h(t)}. \quad (5.7)$$

Moreover, for each integer $q \geq 0$, after decreasing c if necessary,

$$|\partial_t^q \Psi_h(t)| \leq C_q h^{-q/2} (1 + |t|)^q e^{-c S_h(t)}. \quad (5.8)$$

Finally, for any fixed $0 < \delta < 1$,

$$S_h(t) = 3t^2, \quad |t| \leq a_h^{-1}, \quad (5.9)$$

$$S_h(t) \geq c_\delta h^{-1}, \quad |t| \geq \delta a_h^{-1}. \quad (5.10)$$

Proof. For $|u| \leq 1$, the Taylor expansion of $\log \operatorname{sinc}_{\text{rad}} u$ gives $\log |\operatorname{sinc}_{\text{rad}} u| \leq -cu^2$. On $|u| \geq 1$, continuity and $\operatorname{sinc}_{\text{rad}} u \rightarrow 0$ imply $\sup_{|u| \geq 1} |\operatorname{sinc}_{\text{rad}} u| < 1$, proving (5.5). Multiplication over n proves (5.7).

For derivatives, differentiate a finite partial product and use Leibniz's rule. Each differentiation replaces at most one sinc factor by a bounded derivative and contributes a scale $a_h r^n$. The sum of scales is

$$\sum_{n \geq 0} a_h r^n = \frac{a_h}{1-r} = \mathcal{O}(h^{-1/2}).$$

Removing finitely many factors only changes the exponent in (5.7) by a bounded amount. Passing to the infinite product proves (5.8).

If $a_h |t| \leq 1$, every term in (5.6) is quadratic, so (3.4) gives (5.9). If $\delta \leq a_h |t| \leq 1$, then (5.9) and $a_h^2 = 3(1-r^2) \asymp h$ give $S_h(t) \geq c_\delta/h$. If $a_h |t| > 1$, choose the first index after which $a_h r^n |t| < 1$. The quadratic tail alone is bounded below by a constant multiple of $(1-r^2)^{-1} \asymp h^{-1}$. This proves (5.10). $\square$

5.3 Low-frequency cumulant control

Lemma 5.3 (Uniform cumulant remainder). *Fix $R, q \geq 0$ and $0 < \delta < \pi$. Uniformly for $a_h |t| \leq \delta$,*

$$\log \Psi_h(t) = -\frac{t^2}{2} + \sum_{m=2}^{R+1} \frac{\lambda_{2m}(h)}{(2m)!} (it)^{2m} + \rho_{R,h}(t), \quad (5.11)$$

where

$$|\partial_t^q \rho_{R,h}(t)| \leq C_{R,q,\delta} h^{R+1} (1 + |t|)^{2R+4}. \quad (5.12)$$

Proof. For $|z| \leq \delta < \pi$, Taylor's theorem applied to $\log \text{sinc}_{\text{rad}} z$ gives a remainder bounded, together with any fixed number of derivatives, by a constant times $|z|^{2R+4}$. Sum this over $z_n = a_h r^n t$. The omitted power sum is

$$\sum_{n \geq 0} (a_h r^n)^{2R+4} = \frac{a_h^{2R+4}}{1 - r^{2R+4}} = \mathcal{O}(h^{R+1}),$$

because $a_h^2 \asymp h$ and $1 - r^{2R+4} \asymp h$. The retained coefficients are exactly the cumulants in [Proposition 4.1](#). Differentiation only lowers powers of t and changes the constant. $\square$

5.4 Proof of the Edgeworth theorem

Proof of Theorem 5.1. Let

$$A_{R,h}(t) = e^{-t^2/2} \sum_{\substack{\mathbf{k} \in \mathbb{N}_0^R \\ w(\mathbf{k}) \leq R}} \prod_{m=2}^{R+1} \frac{1}{k_m!} \left(\frac{\lambda_{2m}(h)(it)^{2m}}{(2m)!} \right)^{k_m}. \quad (5.13)$$

By (3.7), its inverse Fourier transform is $\varphi \mathcal{E}_{R,h}$.

Choose a small $\varepsilon > 0$, depending on R, j, K , and split the frequency axis into

$$\mathcal{Z}_0 = \{|t| \leq h^{-\varepsilon}\}, \quad \mathcal{Z}_1 = \{h^{-\varepsilon} < |t| \leq \delta/a_h\}, \quad \mathcal{Z}_2 = \{|t| > \delta/a_h\},$$

where $0 < \delta < 1$ is fixed.

On $\mathcal{Z}_0$, [Lemma 5.3](#) applies. Each cumulant term with index m has formal parameter weight $m - 1$, because $\lambda_{2m}(h) = \mathcal{O}(h^{m-1})$. Expanding the exponential in (5.11) and retaining all monomials of total weight at most R produces exactly (5.13). Every omitted monomial has weight at least $R + 1$. If ε is small enough that $h|t|^4 = o(1)$ on $\mathcal{Z}_0$, Taylor's formula for the exponential and (5.12) yield, for every fixed derivative order q ,

$$|\partial_t^q (\Psi_h - A_{R,h})(t)| \leq Ch^{R+1}(1 + |t|)^M e^{-t^2/4} \quad (t \in \mathcal{Z}_0), \quad (5.14)$$

for some $M = M(R, q)$. The right side is integrable with norm $\mathcal{O}(h^{R+1})$.

On $\mathcal{Z}_1$, (5.9) and (5.8) give Gaussian decay e^{-ct^2} . The approximant $A_{R,h}$ is a fixed-degree polynomial times $e^{-t^2/2}$. Therefore every differentiated L^1 norm on $\mathcal{Z}_1$ is

$$\mathcal{O}(e^{-ch^{-2\varepsilon}}) = \mathcal{O}(h^N)$$

for every N .

On $\mathcal{Z}_2$, (5.10) gives an $e^{-c/h}$ factor. For very large t , the number of sinc factors with argument at least one grows like $h^{-1} \log(a_h |t|)$, so (5.7) can be sharpened to an integrable bound

$$|\partial_t^q \Psi_h(t)| \leq C_{q,N} e^{-c/h} (1 + |t|)^{-N}$$

for any prescribed N , once h is small. The Gaussian approximant has the same superpolynomial smallness on $\mathcal{Z}_2$. Hence, for every fixed q ,

$$\|\partial_t^q [(it)^j (\Psi_h - A_{R,h})]\|_{L^1(\mathbb{R})} \leq C_{R,j,q} h^{R+1}. \quad (5.15)$$

Fourier inversion gives the case $K = 0$. Multiplication by x^K in physical space corresponds, after integration by parts, to K derivatives in frequency. Applying (5.15) for all $q \leq K$ proves (5.4). $\square$

5.5 First explicit orders

Expanding the exact-cumulant polynomial in powers of h gives

$$\begin{aligned} f_h(x) = \varphi(x) & \left[1 - \frac{h}{20} H_4(x) \right. \\ & + h^2 \left(\frac{4}{315} H_6(x) + \frac{1}{800} H_8(x) \right) \\ & \left. + h^3 \left(\frac{1}{60} H_4(x) - \frac{3}{700} H_8(x) - \frac{1}{1575} H_{10}(x) - \frac{1}{48000} H_{12}(x) \right) \right] + \mathcal{O}_S(h^4). \end{aligned} \quad (5.16)$$

Here $\mathcal{O}_S$ means an error of that order in every fixed Schwartz seminorm. The second-order terms are the familiar Bell combination

$$\frac{\lambda_6}{6!} H_6 + \frac{1}{2} \left(\frac{\lambda_4}{4!} \right)^2 H_8,$$

but [Theorem 5.1](#) establishes the complete continuous-parameter hierarchy and its global analytic control.

6 CDF and inverse-distribution asymptotics

6.1 Uniform CDF expansion

Because the error in [Theorem 5.1](#) has polynomially weighted uniform control, it is integrable with the same order. Since

$$\int_{-\infty}^x H_n(u) \varphi(u) \, du = -H_{n-1}(x) \varphi(x), \quad n \geq 1, \quad (6.1)$$

we obtain the following.

Corollary 6.1 (All-orders CDF expansion). *Let $F_h(x) = \mathbb{P}(Z_h \leq x)$. For every $R \geq 0$, uniformly in $x \in \mathbb{R}$,*

$$F_h(x) = \Phi(x) - \varphi(x) \sum_{\substack{\mathbf{k} \in \mathbb{N}_0^R \\ 1 \leq w(\mathbf{k}) \leq R}} H_{2\nu(\mathbf{k})-1}(x) \prod_{m=2}^{R+1} \frac{1}{k_m!} \left(\frac{\lambda_{2m}(h)}{(2m)!} \right)^{k_m} + \mathcal{O}(h^{R+1}). \quad (6.2)$$

The error remains $\mathcal{O}(h^{R+1})$ after multiplication by any fixed polynomial weight.

Through second order,

$$F_h(x) = \Phi(x) + \varphi(x) \left[\frac{h}{20} H_3(x) - h^2 \left(\frac{4}{315} H_5(x) + \frac{1}{800} H_7(x) \right) \right] + \mathcal{O}_{\text{weighted}}(h^3). \quad (6.3)$$

6.2 Central Cornish–Fisher inversion

Let $Q_h(u) = F_h^{-1}(u)$, $0 < u < 1$, and $z = \Phi^{-1}(u)$. The inverse Fabius function is the special member $r = 1/2$ after the appropriate affine normalization; the theorem below is an inverse-distribution asymptotic at the opposite, Gaussian boundary $r \uparrow 1$.

Theorem 6.2 (All-orders central quantile expansion). *Fix $0 < \eta < 1/2$ and $R \geq 1$. Uniformly for $u \in [\eta, 1 - \eta]$,*

$$Q_h(u) = z + \sum_{n=1}^R h^n q_n(z) + \mathcal{O}_{R,\eta}(h^{R+1}), \quad z = \Phi^{-1}(u), \quad (6.4)$$

where each q_n is an explicitly computable polynomial. The first two are

$$q_1(z) = -\frac{1}{20}H_3(z), \quad (6.5)$$

$$q_2(z) = \frac{zH_3(z)^2}{800} - \frac{H_4(z)H_3(z)}{400} + \frac{4}{315}H_5(z) + \frac{1}{800}H_7(z). \quad (6.6)$$

All higher q_n follow recursively by coefficient extraction from (6.7) below.

Proof. On the compact interval $u \in [\eta, 1 - \eta]$, the normal quantile z stays in a fixed compact set and $\varphi(z)$ is bounded away from zero. The uniform CDF expansion and the implicit function theorem therefore produce a uniform asymptotic power series in h .

For an explicit recursion, write

$$F_h(x) = \Phi(x) + \varphi(x) \sum_{n \geq 1} h^n A_n(x), \quad q_{<N}(h, z) = \sum_{n=1}^{N-1} h^n q_n(z).$$

Then

$$q_N(z) = -[h^N] \left\{ \frac{\Phi(z + q_{<N}) - \Phi(z)}{\varphi(z)} + \frac{\varphi(z + q_{<N})}{\varphi(z)} \sum_{n=1}^N h^n A_n(z + q_{<N}) \right\}, \quad (6.7)$$

where $[h^N]$ extracts the coefficient. Equivalently, insert $q_{<N} + h^N q_N$ into the identity $F_h(Q_h(u)) = u$ and isolate the unique linear contribution $h^N q_N$. This is a Faà di Bruno/Bell recursion and proves all-orders computability.

From (6.3),

$$A_1 = \frac{1}{20}H_3, \quad A_2 = -\frac{4}{315}H_5 - \frac{1}{800}H_7.$$

At first order, $q_1 = -A_1$, giving (6.5). At second order,

$$q_2 = - \left[\frac{1}{2} \frac{\varphi'}{\varphi} q_1^2 + \frac{(\varphi A_1)'}{\varphi} q_1 + A_2 \right].$$

Use $\varphi'/\varphi = -z$, $(\varphi H_3)' = -\varphi H_4$, and (6.5); simplification gives (6.6). $\square$

Remark 6.3 (Connection to Lagrange inversion). The recursion (6.7) is a moving-base analogue of Lagrange inversion: the base map is Φ , and the perturbation is organized by Hermite polynomials. It provides a concrete meeting point between the repository's inverse-function program, Bell polynomials, and its geometric-comb interpolation program.

7 Moderate deviations and the compact-scale large-deviation principle

7.1 Gaussian moderate deviations

Theorem 7.1 (Moderate-deviation principle). *Let $b_h \rightarrow \infty$ and $hb_h^2 \rightarrow 0$ as $h \downarrow 0$. Then Z_h/b_h satisfies a large-deviation principle with speed b_h^2 and good rate function*

$$I_{\text{mod}}(x) = \frac{x^2}{2}. \quad (7.1)$$

Equivalently, for each Borel set A , the usual LDP upper and lower bounds hold with speed b_h^2 and rate $x^2/2$.

Proof. For fixed t , the scaled log moment generating function is

$$\frac{1}{b_h^2} \log \mathbb{E} e^{tb_h Z_h} = \frac{t^2}{2} + \frac{1}{b_h^2} \sum_{m \geq 2} \frac{\lambda_{2m}(h)}{(2m)!} (tb_h)^{2m}.$$

By Proposition 4.1, $\lambda_{2m}(h) = \mathcal{O}(h^{m-1})$, so the m -th correction is

$$\mathcal{O}((hb_h^2)^{m-1}).$$

The series is locally uniformly controlled because the largest individual moment-generating-function argument is $\mathcal{O}(\sqrt{h} b_h) \rightarrow 0$. Hence the scaled log MGF converges locally uniformly to $t^2/2$. The Gärtner–Ellis theorem [8] gives the LDP with Legendre transform $x^2/2$. $\square$

7.2 The compact scale

Define

$$Y_h = \sqrt{h} Z_h. \quad (7.2)$$

By (1.4), its support converges to $[-\sqrt{6}, \sqrt{6}]$. Put

$$\ell(a) = \log \frac{\sinh a}{a}, \quad \ell(0) = 0, \quad J(a) = \int_0^a \frac{\ell(v)}{v} dv. \quad (7.3)$$

The function J is even, because $\ell(v)/v$ is odd.

Theorem 7.2 (Compact-scale LDP). *The family $(Y_h)_{h \downarrow 0}$ satisfies a large-deviation principle with speed h^{-1} and good convex rate function*

$$I(y) = \sup_{s \in \mathbb{R}} \{sy - \Lambda(s)\}, \quad (7.4)$$

where

$$\Lambda(s) = \int_0^1 \frac{\ell(\sqrt{6} su)}{u} du \quad (7.5)$$

$$= J(\sqrt{6} s). \quad (7.6)$$

The function Λ is even, real analytic, and strictly convex. Its derivative maps $\mathbb{R}$ bijectively onto $(-\sqrt{6}, \sqrt{6})$. Consequently

$$I(y) < \infty \iff |y| < \sqrt{6}, \quad I(y) = \infty \text{ for } |y| \geq \sqrt{6}. \quad (7.7)$$

Proof. The centered summands in Z_h have standardized coefficients $\sqrt{12(1-r^2)}r^n$. Therefore

$$\begin{aligned} h \log \mathbb{E} \exp \left(\frac{s Y_h}{h} \right) &= h \log \mathbb{E} \exp \left(\frac{s Z_h}{\sqrt{h}} \right) \\ &= h \sum_{n \geq 0} \ell \left(b_h s e^{-nh} \right), \quad b_h = \sqrt{\frac{3(1-e^{-2h})}{h}} \rightarrow \sqrt{6}. \end{aligned}$$

The summand is continuous, nonnegative for real arguments, quadratic near zero, and decays exponentially as a function of nh after the argument becomes small. Dominated Riemann-sum convergence gives

$$h \sum_{n \geq 0} \ell(b_n s e^{-nh}) \longrightarrow \int_0^\infty \ell(\sqrt{6} s e^{-x}) \, dx.$$

The substitution $u = e^{-x}$ gives (7.5); the substitution $v = \sqrt{6} su$ gives (7.6).

Each finite- h log MGF is convex, and strict convexity of the limit follows either by differentiating under the integral or by observing that ℓ is the log MGF of a nondegenerate centered uniform variable. Since $\ell(a) = a - \log(2a) + o(1)$ as $a \rightarrow +\infty$,

$$\Lambda'(s) = \sqrt{6} \frac{\ell(\sqrt{6}s)}{\sqrt{6}s} \rightarrow \sqrt{6} \quad (s \rightarrow +\infty),$$

and similarly the limit is $-\sqrt{6}$ at $-\infty$. Thus the derivative range is exactly $(-\sqrt{6}, \sqrt{6})$. The Gärtner–Ellis theorem [8] applies; exponential tightness is automatic from the uniformly bounded supports of Y_h . The Legendre transform is finite only on the derivative range and diverges at and beyond the limiting endpoints, giving (7.7). $\square$

7.3 Exact saddle parametrization

Theorem 7.3 (Parametric rate formula). *For $a > 0$, define*

$$y(a) = \sqrt{6} \frac{\ell(a)}{a}. \quad (7.8)$$

Then $y(a)$ is strictly increasing from 0 to $\sqrt{6}$, and

$$\boxed{I(y(a)) = \ell(a) - J(a)}. \quad (7.9)$$

The negative half of the rate function follows by symmetry.

Proof. Set $a = \sqrt{6}s$. Since $J'(a) = \ell(a)/a$, the saddle equation $y = \Lambda'(s)$ becomes (7.8). Strict convexity of Λ makes $y(a)$ strictly increasing. At the saddle,

$$sy - \Lambda(s) = \frac{a}{\sqrt{6}} \left(\sqrt{6} \frac{\ell(a)}{a} \right) - J(a) = \ell(a) - J(a).$$

$\square$

The Taylor expansion

$$\Lambda(s) = \frac{s^2}{2} - \frac{s^4}{20} + \frac{4}{315}s^6 - \frac{3}{700}s^8 + \frac{16}{9625}s^{10} + \mathcal{O}(s^{12}) \quad (7.10)$$

reuses exactly the Bernoulli coefficients that drive the Edgeworth expansion. Series reversion gives

$$I(y) = \frac{y^2}{2} + \frac{y^4}{20} + \frac{23}{3150}y^6 + \frac{11}{10500}y^8 + \frac{821}{6063750}y^{10} + \mathcal{O}(y^{12}). \quad (7.11)$$

The positive quartic term shows how compact support makes deviations more expensive than Gaussian ones as soon as one leaves the infinitesimal center.

8 Lambert– W inversion at the limiting support edge

The parametric formula in [Theorem 7.3](#) permits a precise edge analysis. It is here that the Gaussian boundary, the lower real branch of Lambert W [7], and the zeta/Stieltjes constants meet.

8.1 A renormalized Mellin constant

Use the standard Stieltjes convention

$$\zeta(1+s) = \frac{1}{s} + \gamma - \gamma_1 s + \mathcal{O}(s^2). \quad (8.1)$$

Define

$$\mathfrak{C} = \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12} = -0.7286939170039306059 \dots \quad (8.2)$$

Lemma 8.1 (Large-saddle expansion of J). *As $a \rightarrow +\infty$, with $L = \log(2a)$,*

$$\ell(a) = a - L + \mathcal{O}(e^{-2a}), \quad (8.3)$$

$$J(a) = a - \frac{1}{2}L^2 + \mathfrak{C} + \mathcal{O}(e^{-2a}/a). \quad (8.4)$$

Proof. The first identity follows from

$$\ell(a) = a - \log(2a) + \log(1 - e^{-2a}).$$

For the second, put $x = 2v$ in (7.3):

$$J(a) = a + \int_0^{2a} \frac{\log(1 - e^{-x}) - \log x}{x} dx. \quad (8.5)$$

The Mellin transform

$$\int_0^\infty x^{s-1} \log(1 - e^{-x}) dx = -\Gamma(s)\zeta(s+1), \quad \Re s > 0, \quad (8.6)$$

continues meromorphically to $s = 0$. Using

$$\Gamma(s) = \frac{1}{s} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12} \right) s + \mathcal{O}(s^2)$$

and (8.1),

$$-\Gamma(s)\zeta(s+1) = -\frac{1}{s^2} + \mathfrak{C} + \mathcal{O}(s). \quad (8.7)$$

More explicitly, the finite part in (8.7) is the absolutely convergent quantity

$$\mathfrak{C} = \int_0^1 \frac{\log(1 - e^{-x}) - \log x}{x} dx + \int_1^\infty \frac{\log(1 - e^{-x})}{x} dx. \quad (8.8)$$

Indeed, for $A > 1$,

$$\int_0^A \frac{\log(1 - e^{-x}) - \log x}{x} dx + \frac{1}{2} \log^2 A = \int_0^1 \frac{\log(1 - e^{-x}) - \log x}{x} dx + \int_1^A \frac{\log(1 - e^{-x})}{x} dx.$$

Taking $A = 2a$ and subtracting (8.8) leaves

$$-\int_{2a}^\infty \frac{\log(1 - e^{-x})}{x} dx = \mathcal{O}(e^{-2a}/a),$$

which proves (8.4). □

8.2 The lower Lambert branch

Let

$$\delta = 1 - \frac{y}{\sqrt{6}}, \quad y \uparrow \sqrt{6}. \quad (8.9)$$

From (7.8),

$$a\delta = \log(2a) - \log(1 - e^{-2a}). \quad (8.10)$$

The exponentially small second term is the sole difference from the exactly Lambert-invertible equation $a\delta = \log(2a)$.

Theorem 8.2 (Lambert- W_{-1} support-edge rate). *Let*

$$w(\delta) = -W_{-1}(-\delta/2), \quad \delta \downarrow 0. \quad (8.11)$$

Then

$$\boxed{I(\sqrt{6}(1 - \delta)) = \frac{1}{2}w(\delta)^2 - w(\delta) - \mathfrak{C} + o(1).} \quad (8.12)$$

More precisely, if a is the exact saddle determined by (8.10), then

$$I(y(a)) = \frac{1}{2}\log^2(2a) - \log(2a) - \mathfrak{C} + \mathcal{O}(e^{-2a}). \quad (8.13)$$

Proof. Subtract (8.4) from (8.3) and use (7.9); this proves (8.13). Ignoring the exponentially small term in (8.10), write $L = \log(2a)$. The equation $a\delta = L$ and $2a = e^L$ give

$$Le^{-L} = \frac{\delta}{2}, \quad L = -W_{-1}(-\delta/2) = w(\delta).$$

For the exact equation, put $L = \log(2a)$ and $F(L) = Le^{-L}$. Multiplying (8.10) by e^{-L} gives

$$\frac{\delta}{2} = F(L) - e^{-L} \log(1 - e^{-e^L}).$$

If $w = w(\delta)$, then $F(w) = \delta/2$. Since $F'(u) = (1 - u)e^{-u}$, the mean-value theorem and $-\log(1 - z) \asymp z$ yield

$$L - w = \mathcal{O}\left(\frac{e^{-e^w}}{w}\right). \quad (8.14)$$

In particular $w(L - w) = o(1)$, so substituting $L = w + \mathcal{O}(e^{-e^w}/w)$ into (8.13) proves (8.12). $\square$

Remark 8.3 (A new three-way connection). The coefficient $\mathfrak{C}$ comes from the finite part of $-\Gamma(s)\zeta(s+1)$ at $s = 0$, the saddle coordinate comes from an infinite geometric uniform convolution, and the edge inversion is controlled by W_{-1} . This is distinct from the fixed- r periodic Lambert- W endpoint expansion in the repository: here $r \uparrow 1$ first, producing a smooth nonperiodic limiting rate, and only then does the compact coordinate approach its support edge.

9 Quadratic q -Gevrey and Denjoy–Carleman regularity

9.1 A convention for global ultradifferentiable classes

Let $M = (M_n)_{n \geq 0}$ be positive with $M_0 = 1$. Define the global Roumieu class

$$\mathcal{B}^{\{M\}}(\mathbb{R}) = \left\{ f \in C^\infty(\mathbb{R}) : \exists C, A > 0 \forall n \geq 0, \left\| f^{(n)} \right\|_\infty \leq CA^n M_n \right\}. \quad (9.1)$$

For log-convex M , the Denjoy–Carleman criterion in this convention [12] says that the class is quasianalytic exactly when

$$\sum_{n \geq 1} \frac{M_{n-1}}{M_n} = \infty. \quad (9.2)$$

A Gevrey class of finite order s corresponds, up to exponential factors, to $M_n = (n!)^s$.

9.2 The Rvachev function

The exact norm identity already established in the repository [14, 17] is

$$\left\| \text{up}^{(n)} \right\|_{\infty} = \left\| \text{up}^{(n)} \right\|_1 = 2^{n(n+1)/2}. \quad (9.3)$$

Its finite signed derivative decomposition is indexed by the signed Thue–Morse sequence [3, 17]. Thus automatic-sequence cancellation and support tiling produce an exact global smoothness scale.

Theorem 9.1 (Sharp Denjoy–Carleman class of up). *Let*

$$M_n^{\text{up}} = 2^{n(n+1)/2}. \quad (9.4)$$

Then:

- (i) $\text{up} \in \mathcal{B}^{\{M^{\text{up}}\}}(\mathbb{R})$, with the sharp constants $C = A = 1$.
- (ii) *The class is optimal modulo the natural exponential equivalence of weight sequences: if $\text{up} \in \mathcal{B}^{\{N\}}$, then $N_n \geq C^{-1} A^{-n} M_n^{\text{up}}$ for suitable $C, A > 0$.*
- (iii) *up belongs to no Gevrey class of finite order.*
- (iv) $\mathcal{B}^{\{M^{\text{up}}\}}$ is nonquasianalytic, since

$$\sum_{n \geq 1} \frac{M_{n-1}^{\text{up}}}{M_n^{\text{up}}} = \sum_{n \geq 1} 2^{-n} = 1. \quad (9.5)$$

Proof. Parts (i) and (ii) are immediate from the equality (9.3). If up were Gevrey of finite order s , then for some C, A ,

$$2^{n(n+1)/2} \leq C A^n (n!)^s.$$

The logarithm of the left side is $(\log 2)n^2/2 + \mathcal{O}(n)$, whereas the right side has logarithm $\mathcal{O}(n \log n)$, a contradiction. Finally

$$M_n^{\text{up}} / M_{n-1}^{\text{up}} = 2^n,$$

so (9.5) and the Denjoy–Carleman criterion prove nonquasianalyticity. $\square$

Definition 9.2 (Quadratic q -Gevrey scale). In this report, a derivative weight with

$$\log M_n = \frac{\log(1/q)}{2} n^2 + \mathcal{O}(n), \quad 0 < q < 1,$$

is called *quadratic q -Gevrey*. This terminology is descriptive and is not intended to replace other established q -Gevrey conventions.

For up, $q = 1/2$. The name records the same quadratic exponent that appears in q -Pochhammer products, Stieltjes–Wigert moments, and log-Gaussian Fourier envelopes.

9.3 Separated-base atomic functions

For the general atomic functions h_a , $a \geq 2$, the repository proves [17]

$$\left\| h_a^{(n)} \right\|_{\infty} = \frac{a^{n(n+3)/2+1}}{2^{n+1}}, \quad \left\| h_a^{(n)} \right\|_1 = a^{n(n+1)/2}. \quad (9.6)$$

Theorem 9.3 (Sharp class for h_a , $a \geq 2$). *Set*

$$M_n^{(a)} = \frac{a^{n(n+3)/2}}{2^n}. \quad (9.7)$$

Then $h_a \in \mathcal{B}^{\{M^{(a)}\}}$ sharply modulo exponential equivalence, belongs to no finite Gevrey class, and its class is nonquasianalytic because

$$\sum_{n \geq 1} \frac{M_{n-1}^{(a)}}{M_n^{(a)}} = \sum_{n \geq 1} \frac{2}{a^{n+1}} < \infty. \quad (9.8)$$

Its quadratic q -Gevrey parameter is $q = a^{-1}$.

Proof. Equation (9.6) is $(a/2)M_n^{(a)}$, proving membership and optimality. Quadratic growth again excludes every finite Gevrey order. A direct calculation gives $M_n^{(a)}/M_{n-1}^{(a)} = a^{n+1}/2$, proving (9.8). $\square$

9.4 Associated functions and Fourier decay

For a weight M , define its associated function

$$\omega_M(t) = \sup_{n \geq 0} \log \frac{t^n}{M_n}. \quad (9.9)$$

For $M_n^{\text{up}} = 2^{n(n+1)/2}$, elementary optimization gives

$$\omega_{M^{\text{up}}}(t) = \frac{(\log t)^2}{2 \log 2} - \frac{1}{2} \log t + \mathcal{O}(1), \quad t \rightarrow \infty. \quad (9.10)$$

Integration by parts and (9.3) therefore imply a log-Gaussian Fourier envelope

$$|\widehat{\text{up}}_{\text{ang}}(-\xi)| \leq C \exp \left[-\frac{(\log(C' |\xi|))^2}{2 \log 2} + \mathcal{O}(\log |\xi|) \right]. \quad (9.11)$$

For h_a , the coefficient $(2 \log a)^{-1}$ replaces $(2 \log 2)^{-1}$. The repository develops the Fourier envelope by product methods; the new point here is that the same quadratic law is precisely the associated function of the optimal Denjoy–Carleman class. The derivative and Fourier pictures are Legendre dual.

10 Numerical experiments

10.1 Method

The exact standardized characteristic function (3.3) was evaluated on a centered FFT frequency grid. The infinite product was truncated only when the largest remaining sinc argument was below 10^{-14} .

Table 2: Successive uniform density errors, with predicted orders h, h^2, h^3, h^4 .

h	Gaussian error	first order	second order	third order
0.30	2.0332×10^{-2}	2.8882×10^{-3}	7.7617×10^{-4}	4.1090×10^{-4}
0.20	1.2983×10^{-2}	1.1427×10^{-3}	1.9456×10^{-4}	6.9283×10^{-5}
0.15	9.5350×10^{-3}	6.1257×10^{-4}	7.3140×10^{-5}	1.8947×10^{-5}
0.10	6.2279×10^{-3}	2.6129×10^{-4}	1.9479×10^{-5}	3.2761×10^{-6}
0.07	4.3071×10^{-3}	1.2518×10^{-4}	6.3096×10^{-6}	7.3484×10^{-7}
0.05	3.0520×10^{-3}	6.2957×10^{-5}	2.2183×10^{-6}	1.8343×10^{-7}

Fourier inversion used 2^{15} grid points with physical spacing 0.002; the numerical density mass was 0.99999999999926 for every reported h . No Monte Carlo sampling was used.

For $E_R(h)$, define the maximum density error on $|x| \leq 8$ after retaining terms through h^R . The results are:

The normalized errors $E_R(h)/h^{R+1}$ remain bounded and settle toward finite values as $h \downarrow 0$, exactly as [Theorem 5.1](#) predicts.

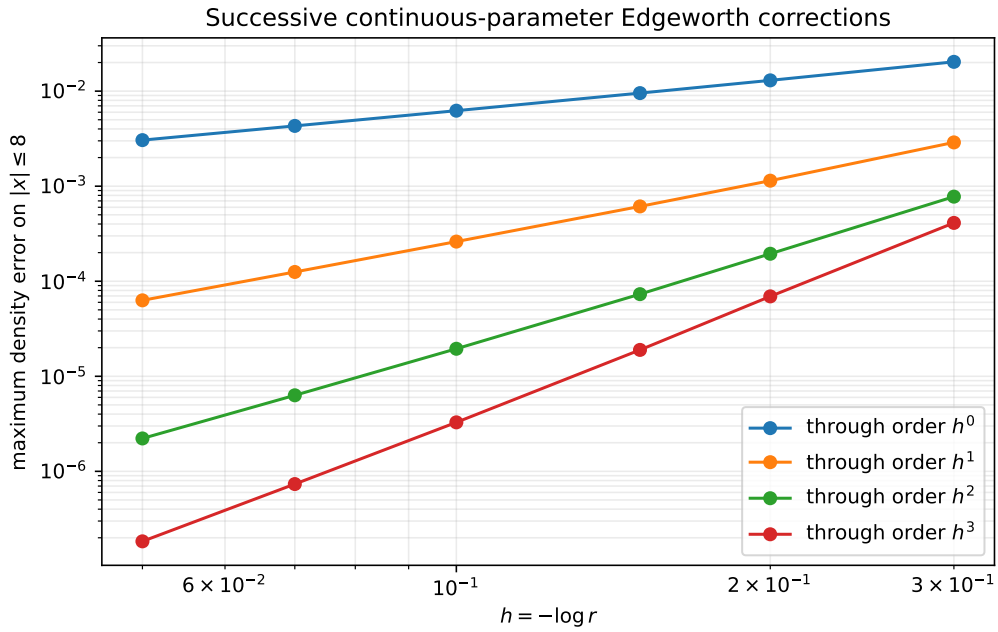


Figure 1: Log-log error curves for the Gaussian approximation and the first three continuous-parameter Edgeworth corrections.

10.2 Central quantiles

At the seven levels $u \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$, the maximum errors are: The second-order error behaves consistently with $\mathcal{O}(h^3)$, while the first-order error behaves as $\mathcal{O}(h^2)$.

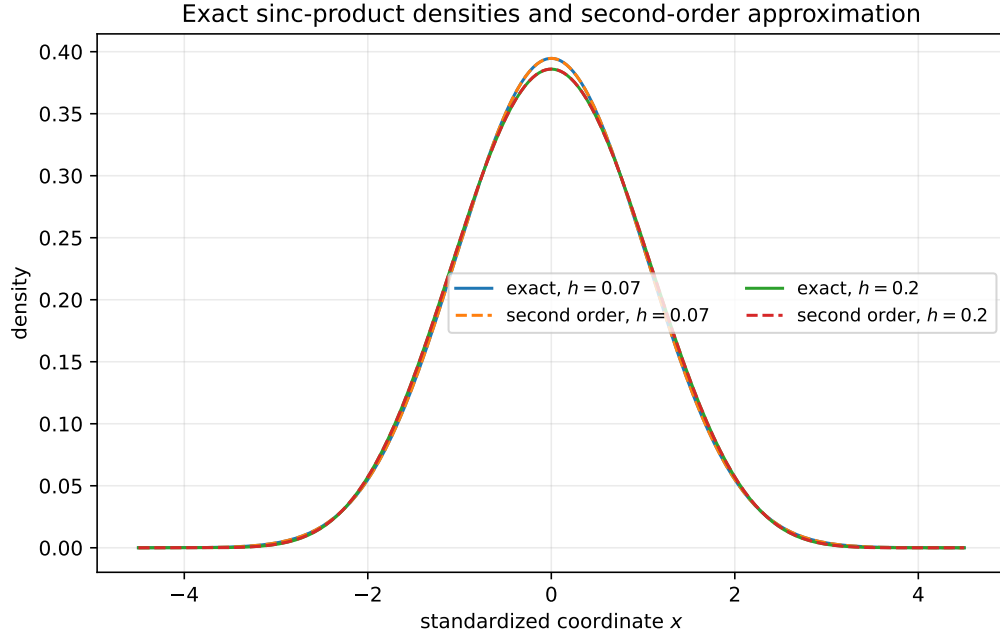


Figure 2: Exact sinc-product densities and the second-order approximation for two values of h . The curves are visually indistinguishable near the center at $h = 0.07$.

Table 3: Central quantile errors for the normal, first-order, and second-order Cornish–Fisher approximations.

h	Gaussian quantile	first order	second order
0.30	3.0365×10^{-2}	4.6160×10^{-3}	2.2050×10^{-3}
0.20	1.9087×10^{-2}	1.9206×10^{-3}	4.8685×10^{-4}
0.15	1.3929×10^{-2}	1.0540×10^{-3}	1.8484×10^{-4}
0.10	9.0414×10^{-3}	4.5827×10^{-4}	4.9114×10^{-5}
0.07	6.2435×10^{-3}	2.2203×10^{-4}	1.5399×10^{-5}
0.05	4.4288×10^{-3}	1.1264×10^{-4}	5.2626×10^{-6}

10.3 Lambert– W edge check

The exact parametric rate (7.9) was compared with the edge form (8.12). Here a is the saddle and $\delta = 1 - y(a)/\sqrt{6}$. The difference is already below 10^{-10} at $a = 10$, despite $\delta \approx 0.30$. This unusually

Table 4: Rapid onset of the Lambert– W support-edge asymptotic.

a	$\delta = 1 - y/\sqrt{6}$	exact $I(y)$	exact minus Lambert– W form
2	0.7023899	0.2810218	-4.246×10^{-3}
3	0.5980804	0.5392933	-3.642×10^{-4}
5	0.4605261	1.0770083	-4.158×10^{-6}
8	0.3465736	1.7997292	-6.640×10^{-9}
10	0.2995732	2.2201676	-9.836×10^{-11}
20	0.1844440	3.8437303	-1.037×10^{-19}

rapid convergence reflects the exponentially small correction in (8.10).

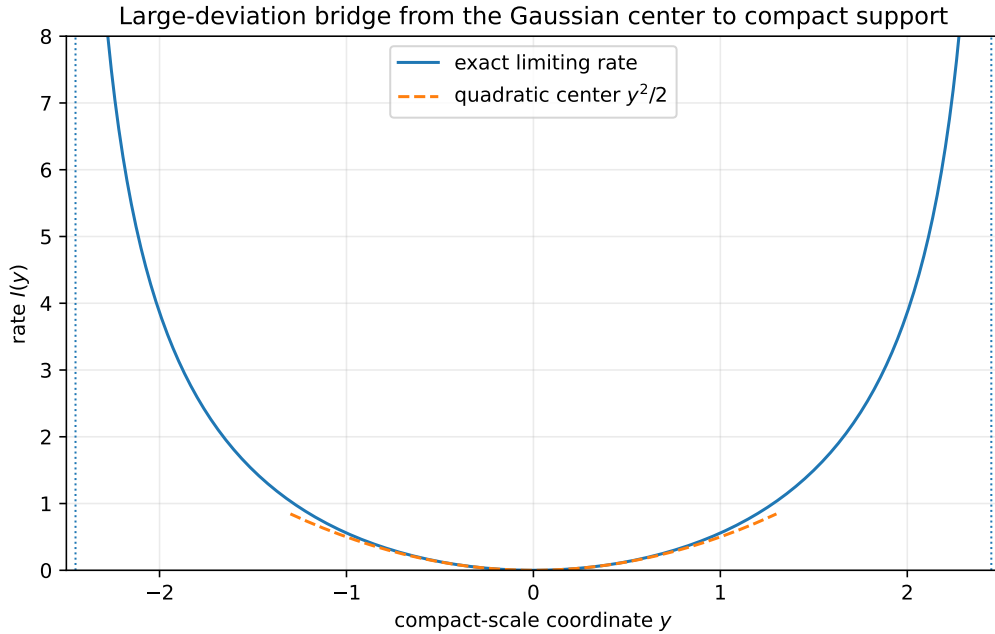


Figure 3: The exact limiting compact-scale rate. It is quadratic near the origin and diverges at the limiting support endpoints $\pm\sqrt{6}$.

11 Conjectures and frontier directions

The theorems above open several directions that are not merely requests for more coefficients. They concern matching between asymptotic regimes, local symbolic regularity, and arithmetic structure.

11.1 A matched four-regime asymptotic theory

Conjecture 11.1 (Uniform saddlepoint local large deviations). *For every compact $K \subset (-\sqrt{6}, \sqrt{6})$, the density of Y_h has the uniform saddlepoint expansion*

$$f_{Y_h}(y) \sim \frac{\exp[-I(y)/h]}{\sqrt{2\pi h \Lambda''(s(y))}} \sum_{n=0}^{\infty} h^n A_n(y), \quad y \in K, \quad (11.1)$$

where $s(y)$ is the unique solution of $\Lambda'(s) = y$, and each A_n is an explicit differential polynomial in Λ .

Conjecture 11.2 (Edgeworth–large-deviation matching). *When $y = \sqrt{h}x \rightarrow 0$, the saddlepoint expansion (11.1), re-expanded in h , matches the Bernoulli–Bell–Hermite Edgeworth series term by term. In the overlap $1 \ll x \ll h^{-1/2}$, it matches the Gaussian moderate-deviation expansion. This matching is Borel summable in a sector of the complex x -plane avoiding the images of the nearest sinc zeros.*

Conjecture 11.3 (Double matching at the compact edge). *There is a two-parameter uniform asymptotic that matches the compact-scale Lambert– W law (8.12) to the fixed- r endpoint asymptotic with its periodic logarithmic profile. The crossover occurs when $\delta \downarrow 0$ and $h \downarrow 0$ jointly so that the exact saddle reaches the first geometric lattice scale before the continuum Riemann-sum approximation loses uniformity.*

The last conjecture should reveal how the periodic fluctuations at fixed r disappear in the continuum $r \uparrow 1$ limit. Poisson summation in the logarithmic index is a promising mechanism: nonzero Fourier modes should be exponentially small in $1/h$ in the interior but become visible near the edge.

11.2 Beyond all orders and sinc-zero geometry

Conjecture 11.4 (Resurgent frequency corrections). *After optimal truncation of the continuous Edgeworth series, the remainder is controlled by complex saddles associated with the nearest zeros of $\text{sinc}_{\text{rad}}(a_h r^n t)$. Their actions form a geometric lattice, and the Stokes multipliers admit a q -product description. The first exponentially small scale is $\exp(-C/h)$, consistent with the outer-frequency estimate in Lemma 5.2.*

A practical route is to apply logarithmic Poisson summation to

$$\sum_{n \geq 0} \log \text{sinc}_{\text{rad}}(a_h e^{-nh} t)$$

and isolate its nonzero dual modes. This should be the Gaussian-boundary counterpart of the periodic endpoint profiles already present at fixed base.

11.3 Entropy and information geometry

The repository conjectures the leading entropy deficit. Theorem 5.1 supplies the missing local expansion, but a complete proof still requires global logarithmic control where f_h/φ is small or large.

Conjecture 11.5 (Full entropic expansion). *The relative entropy has a complete expansion*

$$D(\text{Law}(Z_h) \| N(0, 1)) \sim \sum_{n \geq 2} d_n h^n, \quad d_2 = \frac{3}{100}, \quad (11.2)$$

where every d_n is a finite contraction of Bernoulli-generated Edgeworth coefficients using Hermite orthogonality. The same holds for Fisher information and for Rényi divergences in a nontrivial interval of orders.

A proof should combine the weighted Edgeworth theorem with a saddlepoint majorant on the scale $x = O(h^{-1/2})$. The LDP from Theorem 7.2 provides exactly the missing tail geometry.

11.4 Local symbolic Carleman spectra

The global weight in [Theorem 9.3](#) is sharp, but local derivative growth can be much smaller at points whose symbolic orbit repeatedly enters regions where the base profile is tiny.

Problem 11.6 (Multifractal local regularity). *For $a \geq 2$ and x in the symbolic attractor, determine the spectrum of*

$$\underline{\alpha}(x) = \liminf_{n \rightarrow \infty} \frac{\log |h_a^{(n)}(x)|}{n^2}, \quad \overline{\alpha}(x) = \limsup_{n \rightarrow \infty} \frac{\log |h_a^{(n)}(x)|}{n^2}. \quad (11.3)$$

Relate the Hausdorff dimensions of level sets to recurrence rates of the digit shift and to thermodynamic pressure.

Conjecture 11.7 (Typical quadratic exponent). *For the natural Bernoulli measure on the symbolic attractor of h_a ,*

$$\underline{\alpha}(x) = \overline{\alpha}(x) = \frac{1}{2} \log a$$

for almost every x , while exceptional recurrent orbits realize a nontrivial interval of lower subleading exponents.

11.5 The overlapping phase $1 < a < 2$

The exact norm formula uses disjoint cylinder supports and therefore stops at $a = 2$. Fourier methods still give

$$\log \|h_a^{(n)}\|_{\infty} \leq \frac{\log a}{2} n^2 + \mathcal{O}_a(n \log n).$$

Conjecture 11.8 (Sharp quadratic exponent under overlap). *For every $1 < a < 2$,*

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \log \|h_a^{(n)}\|_{\infty} = \frac{1}{2} \log a. \quad (11.4)$$

The subleading term changes at parameters with exceptional Bernoulli-convolution overlap, and may encode arithmetic properties of a .

This conjecture would extend the quadratic q -Gevrey classification through the entire smooth geometric family, while leaving room for arithmetic phase transitions in the $n \log n$ or linear term.

11.6 Arithmetic and Thue–Morse directions

Conjecture 11.9 (Cumulant-coefficient denominators). *After the natural normalization by $(2m)!$, the coefficients in the h -expansion [\(4.8\)](#) have denominators controlled by von Staudt–Clausen primes from the Bernoulli numbers and by the divisors of $m^{2k} - m$. There is a finite-prime valuation formula compatible with the dyadic arithmetic of $F(2^{-n})$.*

Problem 11.10 (Automatic-sequence Edgeworth analogs). *Replace the uniform digits in [\(1.1\)](#) by signed or block-dependent digits generated by Thue–Morse and other automatic sequences. Determine when a Gaussian boundary survives, how automatic cancellations alter the first nonzero cumulant, and whether the resulting Edgeworth polynomials inherit finite-kernel recurrences.*

Problem 11.11 (Legendre–Hermite contraction). *The repository develops compact-support Legendre and Jacobi expansions, whereas the Gaussian boundary naturally selects the Hermite basis. Determine a joint limit in the polynomial degree and in $h \downarrow 0$ under which the appropriately rescaled Legendre/Jacobi coefficients contract to the Bernoulli–Bell–Hermite Edgeworth coefficients. A successful contraction formula would connect the global compact-support spectral theory to the local Gaussian asymptotic theory without passing through pointwise Fourier inversion.*

11.7 Formalization targets

The analytic arguments naturally split into reusable lemmas suitable for Lean or another proof assistant:

1. exact cumulant algebra and hyperbolic simplification;
2. the global sinc envelope and geometric power-sum bounds;
3. weighted Fourier inversion from $W^{K,1}$ estimates;
4. the Riemann-sum limit for Λ_h ;
5. strict convexity and Legendre duality;
6. the Mellin finite-part computation of $\mathfrak{E}$;
7. the Carleman ratio test for the exact norm sequence.

The most delicate formalization component is the all-orders bookkeeping in [Theorem 5.1](#); encoding the weight $w(\mathbf{k})$ as a graded finite-support map should make the Bell combinatorics manageable.

12 Promising research program

A productive sequence of projects, ordered by mathematical dependency rather than by perceived difficulty, is:

- P1. Local saddlepoint theorem.** Prove [Conjecture 11.1](#) by complex Fourier inversion around the unique imaginary saddle. Establish uniformity on compact subsets of the open limiting support.
- P2. Central matching.** Expand the saddlepoint coefficients at $y = 0$ and identify them combinatorially with the Bernoulli–Bell–Hermite coefficients of [Theorem 5.1](#).
- P3. Entropic CLT with coefficients.** Combine weighted central control and the compact-scale LDP to prove [Conjecture 11.5](#), first through h^3 , then to all orders.
- P4. Logarithmic Poisson modes.** Derive the exponentially small correction between the geometric Riemann sum and its continuum integral. Identify the first dual mode and its Stokes geometry.
- P5. Edge crossover.** Construct the double-scaling asymptotic of [Conjecture 11.3](#), retaining the fixed-base periodic phase until it is exponentially suppressed.
- P6. Overlapping quadratic regularity.** Prove [\(11.4\)](#); investigate whether Pisot, Salem, or exact-overlap parameters alter subleading derivative growth.
- P7. Local Carleman thermodynamics.** Translate the exact symbolic derivative formula into a subadditive pressure problem and compute the multifractal spectrum in [Problem 11.6](#).
- P8. Arithmetic coefficient package.** Implement exact Bernoulli–Bell coefficient generation, valuation analysis, and conjecture discovery for [\(4.8\)](#) and the quantile polynomials.
- P9. Legendre–Hermite contraction.** Derive the double-scaling limit in the preceding problem and use it to transfer spectral estimates between compact-support and Gaussian-boundary regimes.

The first five projects form a single asymptotic arc from the Gaussian center to the fixed-parameter endpoint. Projects six through eight exploit the independent but structurally parallel quadratic growth

appearing in derivatives, q -products, and arithmetic denominators; the ninth connects the repository's orthogonal-polynomial and Gaussian asymptotic languages.

13 Conclusion

The geometric-uniform deformation turns the Fabius/Rvachev pair into a parameterized laboratory in which several asymptotic languages coexist. The exact cumulants make Bernoulli numbers primary; exponential composition turns them into Bell polynomials; Fourier inversion turns Bell monomials into Hermite corrections; inversion of the CDF produces Cornish–Fisher polynomials; rescaling to the compact support produces a Gärtner–Ellis rate; and approaching that support edge produces the lower Lambert branch and a Stieltjes constant. Independently, the Thue–Morse derivative hierarchy produces an exact quadratic-exponential Carleman weight, whose Legendre dual is the log-Gaussian Fourier envelope.

The central result is therefore not only a missing Edgeworth estimate. It is a bridge between the repository's probability, q -series, inverse-function, Lambert– W , Bernoulli/Bell, Fourier-product, and automatic-sequence components. The large-deviation and quadratic q -Gevrey theorems show that the same geometric scaling controls both the parameter boundary $r \uparrow 1$ and the derivative order $n \rightarrow \infty$, but through different Legendre transforms. This duality appears to be a fertile organizing principle for the next phase of the Fabius–Rvachev program.

A Explicit Hermite polynomials and third-order formulas

The probabilists' Hermite polynomials used above begin

$$\begin{aligned} H_3(x) &= x^3 - 3x, \\ H_4(x) &= x^4 - 6x^2 + 3, \\ H_5(x) &= x^5 - 10x^3 + 15x, \\ H_6(x) &= x^6 - 15x^4 + 45x^2 - 15, \\ H_7(x) &= x^7 - 21x^5 + 105x^3 - 105x, \\ H_8(x) &= x^8 - 28x^6 + 210x^4 - 420x^2 + 105. \end{aligned}$$

The third-order CDF expansion obtained by integrating (5.16) is

$$\begin{aligned} F_h(x) = \Phi(x) + \varphi(x) &\left[\frac{h}{20} H_3(x) - h^2 \left(\frac{4}{315} H_5(x) + \frac{1}{800} H_7(x) \right) \right. \\ &\quad \left. + h^3 \left(-\frac{1}{60} H_3(x) + \frac{3}{700} H_7(x) + \frac{1}{1575} H_9(x) + \frac{1}{48000} H_{11}(x) \right) \right] \\ &\quad + \mathcal{O}_{\text{weighted}}(h^4). \end{aligned} \tag{A.1}$$

B A coefficient-extraction form of the full Edgeworth series

Treat h as a formal variable and write the cumulant expansions from Theorem 4.2 as

$$\lambda_{2m}(h) = \sum_{n \geq m-1} \lambda_{2m,n} h^n.$$

Then the coefficient of h^N in the density expansion is

$$P_N(x) = [h^N] \exp_{\mathcal{H}} \left(\sum_{m \geq 2} \frac{\lambda_{2m}(h)}{(2m)!} H_{2m} \right), \quad (\text{B.1})$$

where $\exp_{\mathcal{H}}$ means that multiplication is performed before replacing $(it)^{2k}$ by $H_{2k}(x)$. More explicitly,

$$P_N(x) = \sum_{\substack{k_2, k_3, \dots \geq 0 \\ n_{m,j} \geq 0 \\ \sum_{m,j} j n_{m,j} = N \\ \sum_j n_{m,j} = k_m}} H_{2 \sum_m m k_m}(x) \prod_{m \geq 2} \frac{1}{k_m! (2m)!^{k_m}} \frac{k_m!}{\prod_j n_{m,j}!} \prod_j \lambda_{2m,j}^{n_{m,j}}. \quad (\text{B.2})$$

Only finitely many indices occur for each N , because $\lambda_{2m,j} = 0$ for $j < m - 1$. Formula (B.2) is suitable for exact symbolic generation.

C Large-deviation series derivation

From (4.4),

$$\begin{aligned} \Lambda(s) &= \int_0^1 \sum_{k \geq 1} c_k (\sqrt{6} s u)^{2k} \frac{du}{u} \\ &= \sum_{k \geq 1} \frac{c_k 6^k}{2k} s^{2k}. \end{aligned}$$

This yields (7.10). To compute I , solve $y = \Lambda'(s)$ recursively:

$$s(y) = y + \frac{1}{5}y^3 + \frac{23}{525}y^5 + \frac{22}{2625}y^7 + \frac{821}{606375}y^9 + \mathcal{O}(y^{11}). \quad (\text{C.1})$$

Substitution into $I(y) = ys(y) - \Lambda(s(y))$ gives (7.11).

D Reproducibility and files

The accompanying archive has the following layout:

```
fabius_frontier_report.tex
fabius_frontier_report.pdf
README.md
requirements.txt
MANIFEST.txt
SHA256SUMS
numerics/fabius_q_edgeworth.py
numerics/README.md
data/edgeworth_errors.csv
  data/edgeworth_table.tex
data/quantile_errors.csv
  data/quantile_table.tex
data/large_deviation_edge.csv
  data/ldp_edge_table.tex
figures/edgeworth_error_scaling.{pdf,png}
figures/density_edgeworth_comparison.{pdf,png}
figures/large_deviation_rate.{pdf,png}
```

To install the numerical dependencies and regenerate every CSV table, LaTeX table, and figure:

```
python -m pip install -r requirements.txt
python numerics/fabius_q_edgeworth.py --output-dir .
```

To rebuild the report:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error fabius_frontier_report.tex
```

E Repository source inventory used in the audit

The main live source files and inventory documents consulted were (each canonical file is in its same-named directory):

- FabiusFunction_Mathematical_Glossary.tex;
- Fabius_Function_and_Rvachev_Up.tex;
- Non_Elementarity_of_the_Fabius_Function.tex;
- fabius_lean_walkthrough.tex;
- semi-formalized-research-frontiers.tex;
- the draft MANIFEST.md, with raw inspection of the Gaussian-boundary report, the consolidated exponents-and- q -series volume, the general-base atomic-function chapters, inverse- q reports, Fourier-decay audits, and candidate overlap documents;
- imported LaTeX sources for Arias de Reyna’s compact-support and arithmetic papers, Allouche–Shallit’s Thue–Morse survey, the lacunary-product papers, and the Thue–Morse Dirichlet-series paper;
- the historical archive README and category inventories.

The public path is

[https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/
FabiusFunction/docs](https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs).

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